

Post-Internship Report

A Review of Vision-Language Models and Their Evolution

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13th of September 2025

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1. Executive Summary

This report presents a detailed overview of a two-month research internship undertaken at the Center for Informatics Science (CIS) at Nile University. The primary focus of the internship was on **Vision-Language Models (VLMs)**, a cutting-edge area in artificial intelligence that integrates visual understanding and natural language processing to enable models to interpret and generate descriptions for visual inputs.

The internship provided a structured introduction to applied AI research through a combination of hands-on experimentation, theoretical learning, and academic writing. It began with a practical orientation where we tested and evaluated six different VLMs. This initial phase was designed to familiarize us with various model types, inference pipelines, and dataset preprocessing methods. Through this exploration, we gained experience with model selection, performance evaluation, and the limitations of existing tools.

A significant part of the internship was devoted to an in-depth literature review, through which we explored numerous academic papers related to VLMs, their evaluation strategies, and real-world applications. This helped us build a strong theoretical foundation and allowed us to understand the current research landscape. We also examined papers that focused on different architectural paradigms within VLMs, including encoder-only, decoder-only, and encoder-decoder models, understanding their underlying mechanisms and use cases.

One of the core objectives was to assess the performance of various VLMs on image captioning tasks. We worked with multiple datasets and tested how well each model generated captions in response to different types of images. This allowed us to identify which models were more suitable for generating coherent and semantically accurate descriptions, and which ones struggled with context sensitivity or object recognition. This experimentation phase was essential in helping us understand the strengths and weaknesses of current VLM architectures and their potential for improvement.

The internship culminated in the writing of a comprehensive review paper, where we synthesized our findings into an organized academic report. This paper compared and contrasted different VLMs in terms of their architecture, performance on various tasks, evaluation challenges, and future prospects. It served as both a documentation of our learning journey and a valuable resource for researchers interested in understanding the evolving landscape of vision-language research.

Through this internship, we developed a wide range of technical and research skills. These included working with large-scale datasets, fine-tuning and evaluating AI models, reading and synthesizing scientific literature, and presenting our findings in a coherent and structured format. Beyond technical competencies, the experience enhanced our critical thinking, collaboration, and

academic communication skills, skills that are essential for success in AI research and development.

In conclusion, the research internship at CIS, Nile University was an invaluable experience that allowed us to immerse ourselves in the domain of Vision-Language Models. It provided us with both theoretical grounding and practical experience in applied machine learning and deep learning research, equipping us with the tools and mindset necessary for future contributions to the field.

2. The Project

Our project began with learning how to conduct **systematic search**. We were first introduced to the concept through instructional videos and recommended papers that explained best practices for systematic literature reviews. This initial step helped us establish a structured methodology for exploring the research landscape. As we progressed, we advanced into the study of **machine learning fundamentals** and **large language models (LLMs)**, since these form the foundation for Vision–Language Models (VLMs). We relied on several online resources, including educational YouTube videos, to build this background knowledge.

Once we developed an understanding of LLMs, we moved into the core subject of **Vision–Language Models**. We began by familiarizing ourselves with the basic concepts through video tutorials, followed by a systematic literature search focused on VLMs. During this phase, we identified key research papers addressing **training methods and architectural designs**. A refined search query like “*A Survey on Vision–Language Models: Architectures and Training Approaches*” proved particularly effective. We used platforms such as **Google Scholar, PubMed, and IEEE Xplore**, documenting all relevant findings to ensure reproducibility and clarity in our review process. This research phase laid the groundwork for drafting our academic paper.

On the **technical side**, we applied our taxonomy to select which models to experiment with. Due to hardware constraints, we quickly realized that **large and ultra-large models** require high-end GPUs and were therefore impractical to run on our laptops. Additionally, some medium and small-scale models were not publicly available. Consequently, we focused on models that were both accessible and computationally feasible, namely: **BLIP, GIT, ViT-GPT2, LLaVA, and Qwen-VL**. We also explored **SigLIP and CLIP**, but since it does not include a caption generator, we were unable to produce captions for evaluation.

To implement our experiments, we leveraged multiple resources. The **Hugging Face platform** provided us with a suitable dataset and pre-trained model implementations. Several **GitHub repositories** guided us in applying evaluation metrics, while **YouTube tutorials** offered practical

insights into running and testing models. Furthermore, we benefitted greatly from **regular meetings with domain experts**, whose mentorship helped us overcome technical challenges and refine our approach. Additionally, all our implementation code and experiments are publicly available on GitHub at <https://github.com/MYAR025/Image-Description-using-VLMs->

Through this combination of structured literature review, systematic experimentation, and expert guidance, we were able to produce both a **theoretical review paper** and a **set of empirical results**, bridging research methodology with practical AI experimentation.

2.1 Experiments Overview

To evaluate the performance of Vision–Language Models (VLMs), we carried out a set of experiments using models of different scales. Our team selected three small-scale models, SigLIP, BLIP, and GIT, alongside one medium-scale model, LLaVA 7B. The goal was to understand how model size impacts performance on a shared benchmark and to capture not only quantitative differences but also qualitative behaviors.

Testing is a critical step in working with VLMs because it allows us to validate their ability to interpret images, generate meaningful captions, and stay faithful to the visual content without hallucinating. By running these experiments, we were able to measure the trade-offs between computational efficiency and descriptive accuracy, observe how well each model captures fine-grained details, and assess their fluency and contextual awareness. These insights are key to guiding future improvements and informing decisions about which models are most suitable for different downstream applications.

2.2 Dataset and Experimental Setup

We used the “**jaimin/Image_Caption**” dataset from Hugging Face, which contains roughly fifteen thousand image–caption pairs in a single training split. Each caption is fairly detailed, often describing objects, settings, and actions in depth, making the dataset ideal for testing the generative capabilities of VLMs. Due to limited computational resources, we restricted our experiments to the first 150 rows of the dataset, which still provided a representative sample for evaluating model performance and drawing meaningful insights. This approach preserved the dataset’s diversity while keeping the experiments practical and reproducible.

3. Evaluation Methodology

To assess performance, we employed a comprehensive set of evaluation metrics that capture different aspects of caption quality. The metrics can be grouped into categories

Category	What is measured / what kinds of metrics	Metrics in this group
Traditional n-gram / overlap metrics	Measure surface similarity: how many words or parts of the generated caption overlap exactly (or almost exactly) with ground truth. Good at catching literal matches but less sensitive to paraphrase or meaning.	ROUGE-1 F1, ROUGE-2 F1, ROUGE-L F1; BLEU Score
Lexical and synonym-aware / alignment metrics	These allow matching of synonyms or consider paraphrasing; they look beyond exact string match.	METEOR; BERTScore F1; content word overlap; SPICE-like score; SPICE Alternative (Content)
Consensus-based / human-judgment proxy metrics	Typically compare generated captions to many human references, or leverage weighting of rare words, consensus among human annotations.	CIDEr Score; CIDEr Enhanced (Consensus)
Semantic similarity / embedding-based metrics	Compare meaning rather than word overlap; often use vector embeddings (sentence or word), or learned models trained to approximate human judgments.	Semantic Scores; BERTScore (Contextual); BLEURT Alternative (Learned-like); CLIPScore Alternative (Semantic); MoverScore Alternative (Alignment)
Hybrid / Structural metrics	Consider structure or scene graphs—objects, relationships, attributes—more than just words in isolation.	SPICE-like / SPICE alternatives; sometimes Content Word Overlap; perhaps some parts of the consensus / alignment metrics contribute here

Table.1 “Overview of Metric Types for Caption Quality Evaluation”

4. Results and discussion

4.1 Results:

Model	BLEU	ROUGE-1	ROUGE-L	METEOR	BERTScore
Vit-GPT2	0.2219	0.2347	0.2196	0.2621	0.6304
BLIP	0.00652	0.2179	0.1944	0.1885	0.7820
GIT	0.04254	0.01808	0.01613	0.24713	0.7836
LLaVA	0.8073	0.2153	0.1923	0.0571	0.8701
Qwen-VL	0.3218	0.4237	0.4012	0.3674	0.8345

Table.2“Metric Scores for Image Captioning Models”

Table.2 shows the results of five vision-language models, Vit-GPT2, BLIP, GIT, LLaVA, and Qwen-VL, these models are tested on 150 images using five different evaluation metrics. Qwen-VL scored the highest overall, performing well in both word-based and meaning-based metrics. This means it was able to create captions that closely matched the reference captions in both structure and meaning. LLaVA scored highest in BLEU and BERTScore, showing that its captions were fluent and semantically accurate, but it used less variety in word choice. Vit-GPT2 gave average results across all metrics. In contrast, BLIP and GIT had the lowest scores, especially in metrics that measure word overlap, which suggests their captions were quite different from the original ones. Overall, the results show that newer models like Qwen-VL and LLaVA are better at generating high-quality image captions.

BERTScore was higher for most models because it doesn't need exact word matches. Instead, it checks if the meaning of the caption is similar to the reference by using word embeddings. This allows models to get a good score even if they use different words, as long as the overall meaning is correct.

4.2 Discussion

The evaluation results clearly show differences in how well each model performs on the image captioning task. Qwen-VL stood out with the most consistent and high scores across all metrics, showing that it can understand image content and describe it in a way that matches both the wording and meaning of human-written captions. Its strong results in both ROUGE and METEOR suggest that it balances lexical overlap with semantic understanding, which is important for generating captions that are both accurate and natural.

LLaVA also performed strongly, especially in BLEU and BERTScore, meaning it generates fluent and semantically meaningful captions. However, its lower scores in METEOR and ROUGE suggest that it may rely more on repetitive or common phrasing, rather than using

varied and expressive language. This shows that while LLaVA understands the general idea of the image, it may be less flexible in how it describes it. In contrast, Qwen-VL scored higher in ROUGE and METEOR because these metrics reward models that use more varied words and capture more of the important content, even if the wording is different. This means Qwen-VL uses more flexible language and includes more key ideas from the reference captions.

Vit-GPT2 showed moderate performance. Although it is less advanced than the other models, it still produced acceptable results across all metrics. This suggests that with the right setup, simpler models can still be useful, especially in situations where resources are limited.

On the other hand, BLIP and GIT had the weakest scores in most metrics. These models often produced captions that were either too general or lacked alignment with the reference captions. Their poor performance in overlap-based metrics like BLEU and ROUGE shows that their outputs differ significantly from human-written captions in both wording and structure. This could be due to differences in training data or limitations in how these models process visual information.

5. Conclusion

This internship offered a comprehensive exploration of Vision–Language Models, combining theoretical study with practical experimentation. Through systematic literature review and hands-on testing of models like BLIP, GIT, ViT-GPT2, LLaVA, Qwen-VL, and SigLIP, we gained insight into model architecture, training methods, and evaluation strategies.

Our experiments highlighted the strengths and limitations of each model, showing how larger, advanced models produce more accurate and meaningful captions, while smaller models face challenges in alignment and detail. Beyond technical knowledge, the experience enhanced our research, critical thinking, and communication skills.

Overall, this project provided a strong foundation for further AI research and development, and all implementation code is publicly available on GitHub for transparency and future exploration.

6.Samples

We evaluated our image dataset using five different models: GIT, BLIP, Vit-GPT2, QWEN and LLaVA. Each model generated captions for the images, demonstrating varying levels of detail and accuracy. The captions produced by the models vary in focus, capturing aspects such as objects, colors, and context. These are 5 captions out of 150 generated in total, highlighting how different models emphasize different elements of the images.

Dataset pictures with ground truth	 : “In this picture we can see people driving old cars on the ground covered with mud and surrounded by barricades, metal poles, fence, divider, wooden fence, poles etc., We can see some text and numbers on the cars. In the background, we can see grass, poles and other vehicles on the ground.”	 “In this image we can see a statue with blue coat and white shirt in background we can see so many flowers, so many trees, and a clear sky.”	 “At the bottom of the picture, we see the bricks, stones, cardboard boxes, bucket, white color boxes, wooden sticks, black color plastic cover, black color tray like object. On the right side, we see the stairs and the railing. We even see the monitor and some electronic goods. Behind that, we see a building and a creeper plant. Beside that, we see a small house and a black color cover. On the left side, we see a tree and a white color building.”	 “In this image we can see a spherical object and other objects on the ground. There are few persons standing and a person is walking on the ground. In the background we can see plants and trees.”	 “In this image, we can see some trees and buildings. There is a hanging flower pot on the right side of the image. There is a safety barrier in the middle of the image. There is a car on the road. There are some plants on the left side of the image. There are clouds in the sky. There is a pole at the top of the image.”
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Vit-gpt2 (Small)	a car is being driven by a group of people	a man dressed in a costume holding a bird	a pile of trash sitting on the side of a building	a man and a little girl playing with a frisbee	a flower arrangement in the window of a building
Blip (Small)	a group of cars racing around a dirt track	“a woman in a blue dress and hat standing in a field”	“a pile of junk”	“a man and a little boy playing with a ball”	“a balcony with a plant on it”
GIT (Small)	a black car with yellow numbers	“a blue vest”	“the building is made of brick”	“the grass is green.”	“a wall on the side of a building”
LLaVa (Medium)	"In this picture we can see people driving old cars on the ground covered with mud and surrounded by barricades, metal poles, fence, divider, wooden fence, poles etc., We can see some text and numbers on the cars. In the background, we can see grass, poles and other vehicles on the ground.", "The image captures a thrilling scene of two cars racing on a dirt track. The cars are positioned close to each other, with one car	In this image we can see a statue with blue coat and white shirt in background we can see so many flowers, so many trees, and a clear sky.", "The image features a garden scene with a wooden dummy dressed as a woman, standing in the middle of a field of flowers. The dummy is wearing a blue dress and has a crab on its	"At the bottom of the picture, we see the bricks, stones, cardboard boxes, bucket, white color boxes, wooden sticks, black color plastic cover, black color tray like object. On the right side, we see the stairs and the railing. We even see the monitor and some electronic goods. Behind that, we see a building and a creeper plant. Beside that, we see a small house and a black color cover. On the left side, we see a tree and a white color building.", "The image depicts a dilapidated house with a large pile of	In this image we can see a spherical object and other objects on the ground. There are few persons standing and a person is walking on the ground. In the background we can see plants and trees.", "The image features a group of people, including both adults and children, playing with a red ball in a grassy yard. The group is spread out across the yard, with some people standing closer to the ball while others are further away. There are several sports balls	"In this image, we can see some trees and buildings. There is a hanging flower pot on the right side of the image. There is a safety barrier in the middle of the image. There is a car on the road. There are some plants on the left side of the image. There are clouds in the sky. There is a pole at the top of the image.", "The image features a balcony with a flower pot filled with pink flowers, likely

	slightly ahead of the other. The cars are in motion, kicking up dirt as they speed along the track. There are several people in the scene, likely spectators or participants in the race. Some of them are standing near the cars, while others are scattered around the track. The atmosphere is lively.	back. The garden is filled with various flowers, creating a vibrant and lively atmosphere. In addition to the wooden dummy, there are two other people in the scene, one standing near the left edge of the image and the other person located towards the right side. A handbag"	junk and debris in front of it. The pile consists of various items, including multiple TVs, a refrigerator, and a microwave. The scene also features a collection of old appliances and a pile of bricks. In addition to the junk, there is a window on the side of the house, and a staircase can be seen"	scattered around the yard, indicating that the group is engaged in a fun and active game. The scene captures the joy and excitement of the participants as they enjoy their time together outdoors."	roses. The flowers are hanging over the railing, creating a beautiful and colorful display. In the background, there are several cars parked on the street, with one car closer to the left side of the image and two more cars on the right side. Additionally, there are two people visible in the scene, one near the left edge of the image and the other closer to"
Qwen-VL (Medium)	"The image depicts a scene from a demolition derby, a type of motorsport event where participants drive modified	"The image depicts a whimsical, handmade scarecrow standing in a field of vibrant yellow flowers. The scarecrow is	"The image depicts an exterior view of an old, dilapidated building. The building appears to be in a state of disrepair, with visible signs of neglect and abandonment. The	"The image depicts a scene in a well-maintained garden or backyard. The primary focus is on a group of four individuals, three adults and one child, standing on a lush, green	"The image depicts a balcony view from an apartment building. The balcony is enclosed with white metal

	cars into each other and into barriers in an attempt to cause damage to their opponents' vehicles. The setting appears to be an outdoor dirt track, likely part of a fairground or a dedicated demolition derby venue.”	dressed in a blue apron with a red crab hanging from the front pocket. The apron is tied around the waist with a blue string. The scarecrow is wearing a white shirt and a white hat adorned with a pink shell. The hat appears to be made of straw or a similar material. The scarecrow's arms are outstretched, holding a wooden fork with a handle”	structure is made of weathered wooden planks, and the walls are covered with patches of moss and ivy, indicating prolonged exposure to the elements. The windows are broken or boarded up, and the overall condition of the building suggests it has been unused for a significant period.	lawn. The lawn is neatly trimmed, with a vibrant red ball placed on the grass near the bottom left corner of the image. The red ball appears to be a large, inflatable ball, possibly used for outdoor games or activities.	railings, which are visible in the foreground. Hanging from the railing is a flowerpot containing a plant with bright red flowers. The flowers are in full bloom, adding a vibrant touch to the scene.
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Table.3 “Example Captions and Answers Generated by VLMs”

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